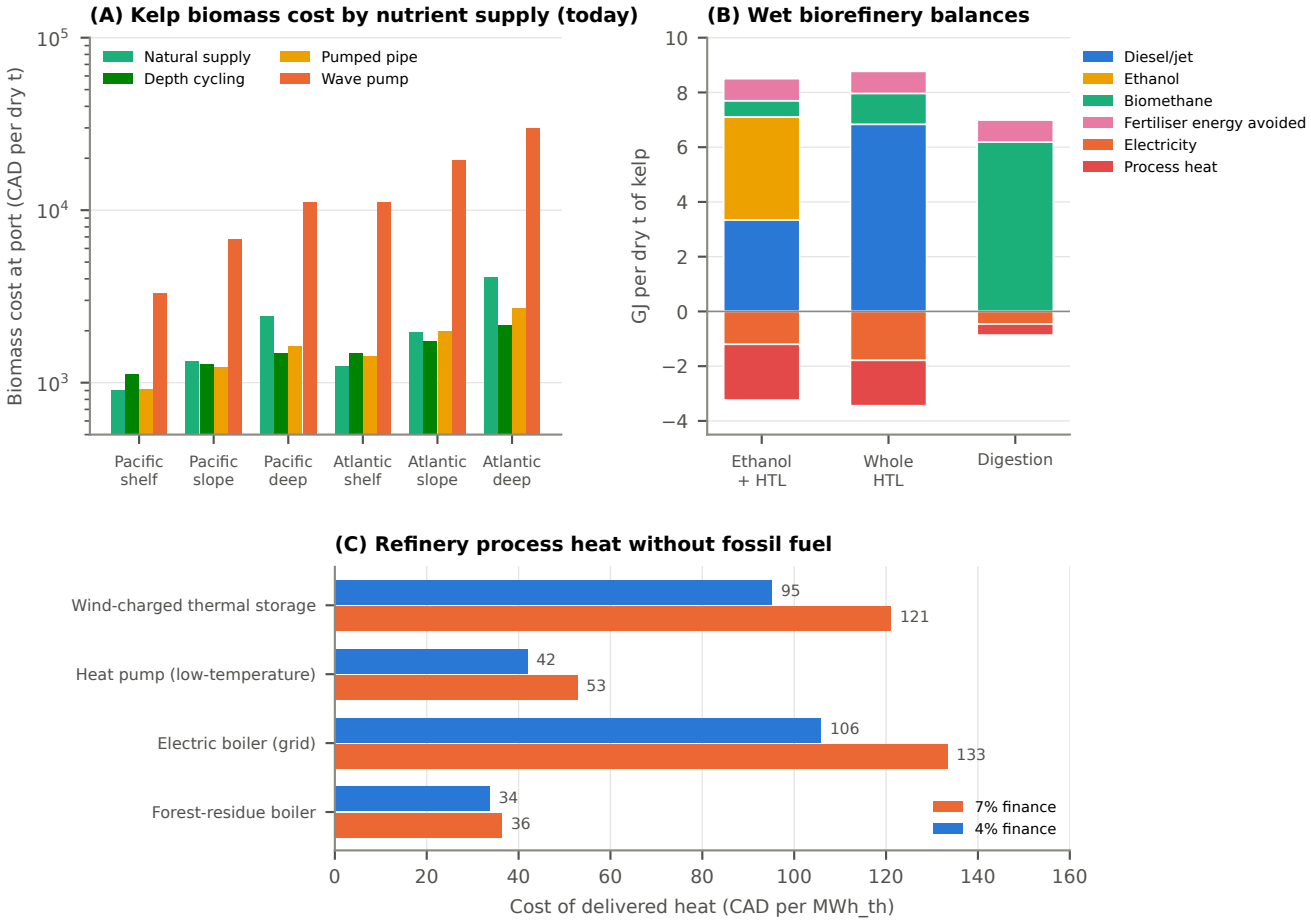
(A) Kelp biomass cost by nutrient supply (today)
Biomass cost at port (CAD per dry t)
Natural supply
Depth cycling
Pumped pipe
Wave pump
Pacific shelf
Pacific slope
Pacific deep
Atlantic shelf
Atlantic slope
Atlantic deep
(B) Wet biorefinery balances
GJ per dry t of kelp
Diesel/jet
Ethanol
Biomethane
Fertiliser energy avoided
Electricity
Process heat
Ethanol + HTL
Whole HTL
Digestion
(C) Refinery process heat without fossil fuel
Wind-charged thermal storage
Heat pump (low-temperature)
Electric boiler (grid)
Forest-residue boiler
95
121
42
53
106
133
34
36
7% finance
4% finance
Cost of delivered heat (CAD per MWh_th)